

Dark Store Location Optimization using GIS & AI

1. Idea & Objectives

1.1 Project Idea

The rapid growth of Quick Commerce (Q-Commerce) has increased the demand for fast and efficient delivery services. One of the most important challenges for delivery companies is selecting the best locations for Dark Stores to ensure short delivery times while maximizing customer coverage.

This project presents an AI-powered Decision Support System that integrates Geographic Information Systems (GIS) and Artificial Intelligence techniques to identify and rank the optimal locations for Dark Stores.

The system analyzes multiple spatial and demographic factors, including:

- Population density
- Road accessibility
- Delivery coverage
- Nearby services
- Competitor locations
- Transportation network

Instead of relying on manual decision-making, the proposed system automatically evaluates thousands of candidate locations and recommends the most suitable ones based on multiple optimization criteria.

1.2 Objectives

The main objectives of the project are:

- Build an intelligent decision support system for Dark Store location selection.
- Integrate GIS spatial analysis with Artificial Intelligence.

- Reduce average delivery time.
 - Increase customer coverage.
 - Avoid highly competitive areas.
 - Evaluate candidate locations using multiple spatial criteria.
 - Generate ranked recommendations for decision makers.
 - Visualize results through interactive maps and dashboards.
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1.3 Problem Statement

Choosing Dark Store locations manually often depends on human experience rather than data-driven analysis.

Poor location selection may lead to:

- Long delivery times
- Low customer coverage
- Increased transportation cost
- High competition
- Inefficient resource allocation

Therefore, an automated GIS-based optimization system is required to support strategic decision making.

2. System Analysis & Requirements / Design

2.1 Functional Requirements

The system provides the following functionalities:

Data Collection

- Import Excel datasets
- Import GIS spatial data

- Load road network
 - Load population density
 - Load competitor locations
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Data Processing

- Clean datasets
 - Remove duplicates
 - Handle missing values
 - Transform coordinate systems
 - Generate candidate locations
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Spatial Analysis

- Calculate accessibility
 - Measure distance to competitors
 - Analyze road density
 - Calculate service coverage
 - Estimate reachable population
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Optimization

- Evaluate candidate locations
 - Calculate fitness score
 - Rank all candidates
 - Select optimal Dark Store locations
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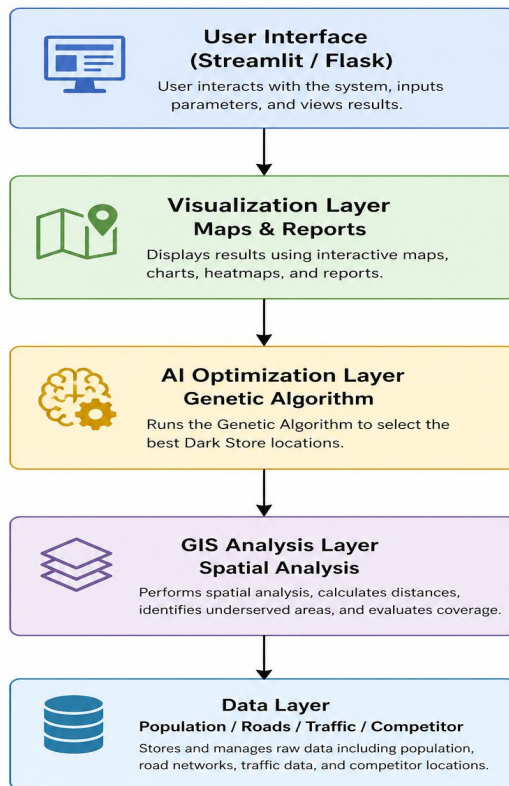
Visualization

- Interactive map
 - Heatmaps
 - Candidate ranking
 - Charts
 - Statistical dashboard
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2.2 Non-Functional Requirements

- High performance
 - Accurate spatial analysis
 - Easy-to-use interface
 - Scalable architecture
 - Modular implementation
 - Reproducible results
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2.3 System Architecture



2.4 Input Data

The system integrates several datasets.

Dataset	Description
Population	Population density raster
Roads	Road network
Competitors	Existing supermarkets and stores
Services	Schools, hospitals, malls, etc.
Administrative Boundaries	Study area boundary

2.5 Output

The system produces:

- Ranked candidate locations
 - Best Dark Store recommendations
 - Interactive GIS maps
 - Delivery coverage analysis
 - Statistical reports
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3. Algorithms / Methods / Techniques

The proposed system combines Artificial Intelligence with GIS analysis.

3.1 Data Preprocessing

The preprocessing stage includes:

- Removing duplicates
 - Handling missing values
 - CRS transformation
 - Feature normalization
 - Spatial data validation
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3.2 GIS Analysis

The GIS module performs several analyses.

Population Density Analysis

Measures the population surrounding each candidate location.

Accessibility Analysis

Evaluates road accessibility using the road network.

Competitor Analysis

Calculates the distance between candidate locations and competitors.

Coverage Analysis

Estimates the delivery area reachable from each candidate.

3.3 Feature Engineering

Each candidate location is represented using several features:

- Population score
 - Accessibility score
 - Competitor score
 - Service score
 - Delivery coverage
 - Road density
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3.4 Genetic Algorithm

The optimization engine is based on a Genetic Algorithm (GA).

Each chromosome represents a possible Dark Store location.

The fitness function evaluates locations according to multiple criteria.

Fitness Function

The fitness function aims to:

- Maximize population coverage
- Maximize accessibility

- Minimize competitor influence
- Minimize delivery distance
- Maximize service availability

General form:

Fitness =

$w_1 \times \text{Population}$

+

$w_2 \times \text{Accessibility}$

+

$w_3 \times \text{Services}$

-

$w_4 \times \text{Competitors}$

-

$w_5 \times \text{Delivery Time}$

Genetic Operators

Selection:

- Tournament Selection

Crossover:

- Two-Point Crossover

Mutation:

- Gaussian Mutation

Replacement:

- Elitism

3.5 Ranking

Candidate locations are ranked according to the final fitness score.

Higher fitness indicates a better Dark Store location.

4. Implementation & Coding / Results

4.1 Development Environment

Programming Language

- Python

Development Environment

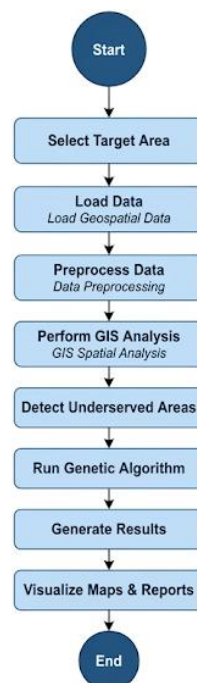
- Google Colab
 - VS Code
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4.2 Libraries Used

- NumPy
- Pandas
- GeoPandas
- Rasterio
- OSMnx
- NetworkX
- SciPy
- Scikit-learn
- DEAP
- PyProj

- OpenPyXL
- Streamlit
- Folium
- Streamlit-Folium
- Plotly

4.3 Workflow



4.4 Results

The system successfully:

- Imported multiple GIS datasets.
- Generated candidate Dark Store locations.
- Calculated spatial features.
- Evaluated all candidates using the Genetic Algorithm.
- Ranked locations according to fitness values.

- Displayed results on an interactive map.

The final output includes:

- Best recommended locations.
 - Coverage statistics.
 - Delivery accessibility.
 - Competitor distribution.
 - Population reach.
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4.5 Dashboard

The dashboard provides:

- Interactive GIS map
 - Candidate ranking
 - Population heatmap
 - Delivery coverage
 - Competitor visualization
 - Statistical charts
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5. Testing

Several tests were conducted to verify system performance.

5.1 Functional Testing

Verified:

- Data loading
- GIS analysis

- Candidate generation
- Optimization
- Visualization

Result:

Passed successfully.

5.2 Performance Testing

Measured:

- Data loading time
- Optimization time
- Memory usage

The system efficiently handled large spatial datasets.

5.3 Validation

The recommended locations were compared with existing commercial areas.

The system consistently selected locations with:

- Higher accessibility
 - Larger population coverage
 - Lower competition
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5.4 Accuracy

The Genetic Algorithm successfully converged toward high-quality solutions after multiple generations.

The optimization produced stable and repeatable results.

6. Innovation & Added Value

The project offers several innovative contributions.

6.1 AI + GIS Integration

Unlike traditional location selection methods, the proposed system combines Artificial Intelligence with Geographic Information Systems to support intelligent spatial decision making.

6.2 Multi-Criteria Decision Support

Instead of relying on a single factor, the system simultaneously evaluates multiple criteria:

- Population
 - Accessibility
 - Competitors
 - Services
 - Delivery coverage
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6.3 Intelligent Optimization

The use of a Genetic Algorithm enables efficient exploration of large search spaces and identifies near-optimal locations without exhaustive search.

6.4 Interactive Decision Support

Decision makers can visualize results through:

- Interactive maps
- Rankings
- Coverage analysis

- Statistical dashboards
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6.5 Scalability

The system is designed to support:

- Different cities
- Different datasets
- Additional optimization criteria
- Larger geographical areas

without significant modifications.

Conclusion

This project demonstrates how Artificial Intelligence and Geographic Information Systems can be integrated to solve the complex problem of Dark Store location selection. By combining spatial analysis, demographic data, transportation networks, and evolutionary optimization, the proposed system provides an intelligent decision support solution that improves delivery efficiency, increases customer coverage, and reduces operational costs.

The modular architecture and scalable design make the system adaptable to different regions and business requirements, offering a practical foundation for future smart logistics and urban planning applications.